Input, Process, Output

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Input, Process, Output before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

AI or Not?

Without opening the old lesson, explain one key idea from AI or Not?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Input, Process, Output trace and record the first step you can explain without help.

Use W2: Finish the visitor example for **Representation**s and information paths. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Input, Process, Output explanation against the supplied facts before discussing it.

Use the worked case for **Representation**s and information paths. Proposed explanation: “Every **output** is the final action”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Investigate a new loss of information

A classroom device receives four scores and stores only their total, 20. A student claims that every score must be 5. Give two different lists of four nonnegative whole-number scores with total 20.

Explain precisely what the stored total establishes and what it does not.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Input, Process, Output target to a trace and a justified conclusion.

A library kiosk receives 'mystery', filters a catalogue by genre and displays three titles. No speaker is attached. Draw its information path. A visitor says it read three books aloud. Which stage contradicts that claim?

Solve the given case

> Solve the supplied Input, Process, Output case and show the deciding evidence.

A playlist helper receives a mood, matches song tags and displays three titles.

Name **input**, **process** and **output**. Is displaying titles the same as playing audio?

Show your **trace**, calculation or **evidence**

Write or draw your reasoning and evidence.

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A classroom fan receives a temperature reading, compares it with 25 degrees C, and turns on. Name the **input**, **process** and **output**. What should happen if the reading is missing?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 2 Test and reflect

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	

Test record	My evidence
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Input, Process, Output adds to Helpful Classroom Sorter and keep the evidence.

Draw the Classroom Sorter's first **input-process-output** map and show where a person may review the result.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.

- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A library kiosk receives 'mystery', filters a catalogue by genre and displays three titles. No speaker is attached. Draw its information path. A visitor says it read three books aloud. Which stage contradicts that claim?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Trace a request

> Trace a request. Apply the supplied Input, Process, Output facts and keep your reasoning.

The kiosk receives genre=science and maximum pages=120. The catalogue contains A science 80 pages, B science 150 pages and C mystery 90 pages. It keeps records with the requested genre and no more than 120 pages.

Draw the **input**, **process** and **output**, naming the surviving record.

> **My reasoning and evidence**

W2 Finish the visitor example

> Finish the visitor example. Apply the supplied Input, Process, Output facts and keep your reasoning.

A record stores only total visitors=3. Give two different arrival schedules that produce this total. Can the total tell us the busiest hour? Explain.

> **My reasoning and evidence**

WORKBOOK · 14 OF 21

W3 Choose a sufficient representation

> Choose a sufficient representation. Apply the supplied Input, Process, Output facts and keep your reasoning.

For each task, choose the information needed: decide whether a book is under 120 pages; quote its first sentence exactly; decide whether a copy is available now. Explain why a genre tag alone is insufficient for all three.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Investigate a new loss of information

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Investigate a new loss of information. Apply the supplied Input, Process, Output facts and keep your reasoning.

A classroom device receives four scores and stores only their total, 20. A student claims that every score must be 5. Give two different lists of four nonnegative whole-number scores with total 20.

Explain precisely what the stored total establishes and what it does not.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Diagnose before repairing

> Diagnose before repairing. Apply the supplied Input, Process, Output facts and keep your reasoning.

A voice calculator receives a spoken request for 16 divided by 4. Its transcript says 60 divided by 4 and its calculator returns 15.

Identify the first incorrect stage, show the intended result and propose a check before calculation.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Input, Process, Output

> A fictional classroom cupboard records coloured pencils, then reports just a total. Explain the labelled step before checking the changed condition.

A fictional classroom cupboard records coloured pencils, then reports just a total.

Box A has 2 red and 4 blue pencils. Box B has 5 red and 1 blue.

The **process** adds colour counts and stores only the total, not the two counts.

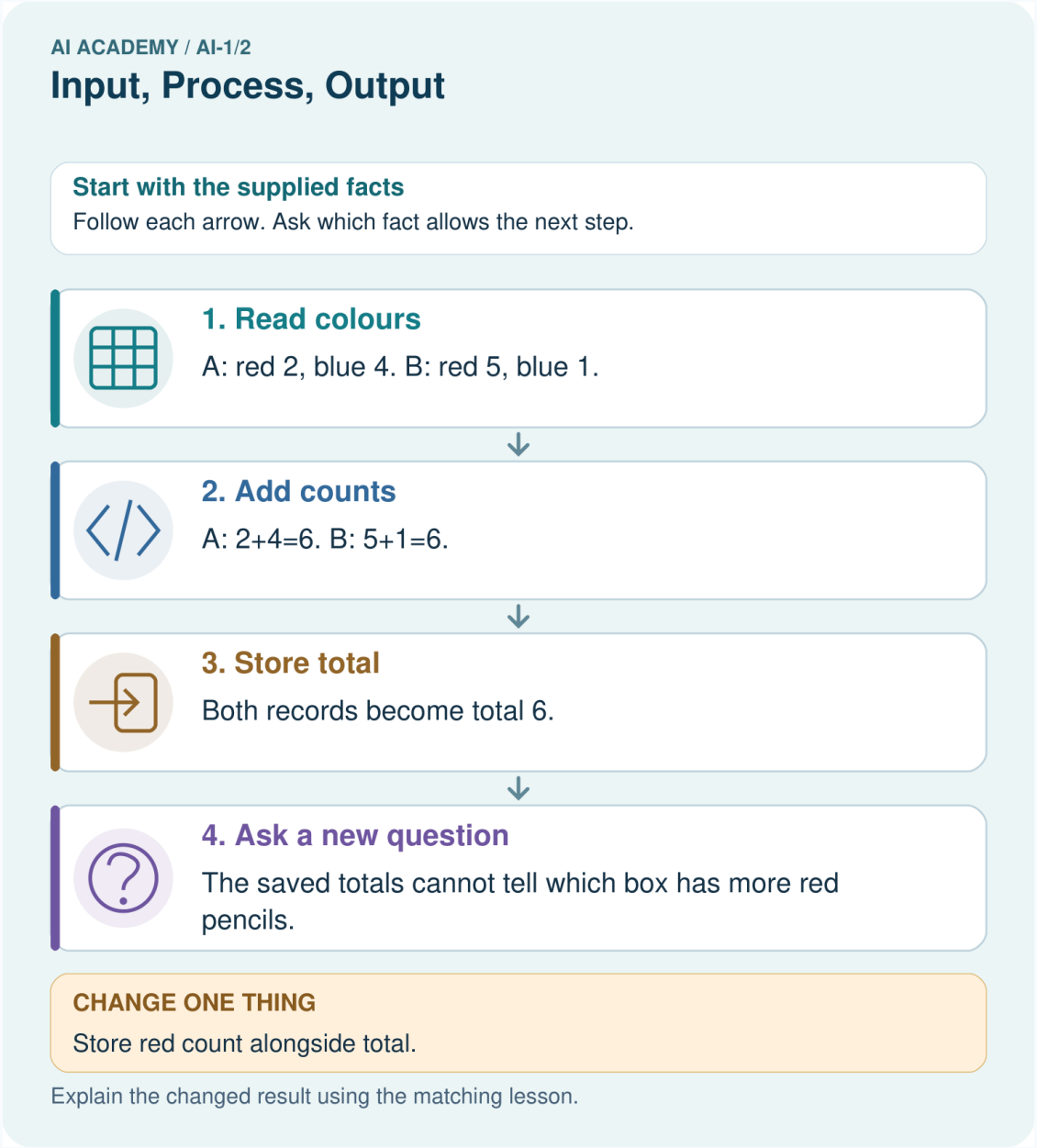
AI ACADEMY / AI-1/2

Different colour counts, same total

	Red	Blue	Total
Box A	2	4	6
Box B	5	1	6

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Use the richer record to find blue pencils in Box B. Explain why total alone was insufficient.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Choose sorter **input**s that retain the clues needed for its decision; draw the transformation and identify discarded details.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Choose sorter **input**s that retain the clues needed for its decision; draw the transformation and identify discarded details.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional classroom cupboard records coloured pencils, then reports just a total.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The saved totals cannot tell which box has more red pencils.

1. Add counts

2. Read colours

3. Ask a new question

4. Store total

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Ask a new question The saved totals cannot tell which box has more red pencils.

Adding preserves the number of pencils but discards how that number was divided between colours. Both distinct **input**s lead to the same stored **output**.

Working backwards from 6 therefore has several possible answers; it is not an instruction to divide equally.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Store red count alongside total. Which conclusion is supported by the supplied facts?

1. The records become (6,2) and (6,5).

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial **trace**. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The records become (6,2) and (6,5). Blue count can now be recovered by total minus red, assuming these are the only colours.

Repair a missing building block

Review AI or Not? (ai-1/1). Goal: Distinguish an observed result from an explanation of how it was produced. Short practice: Point to the learned stage. Does the changed display prove the weather forecast changed?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

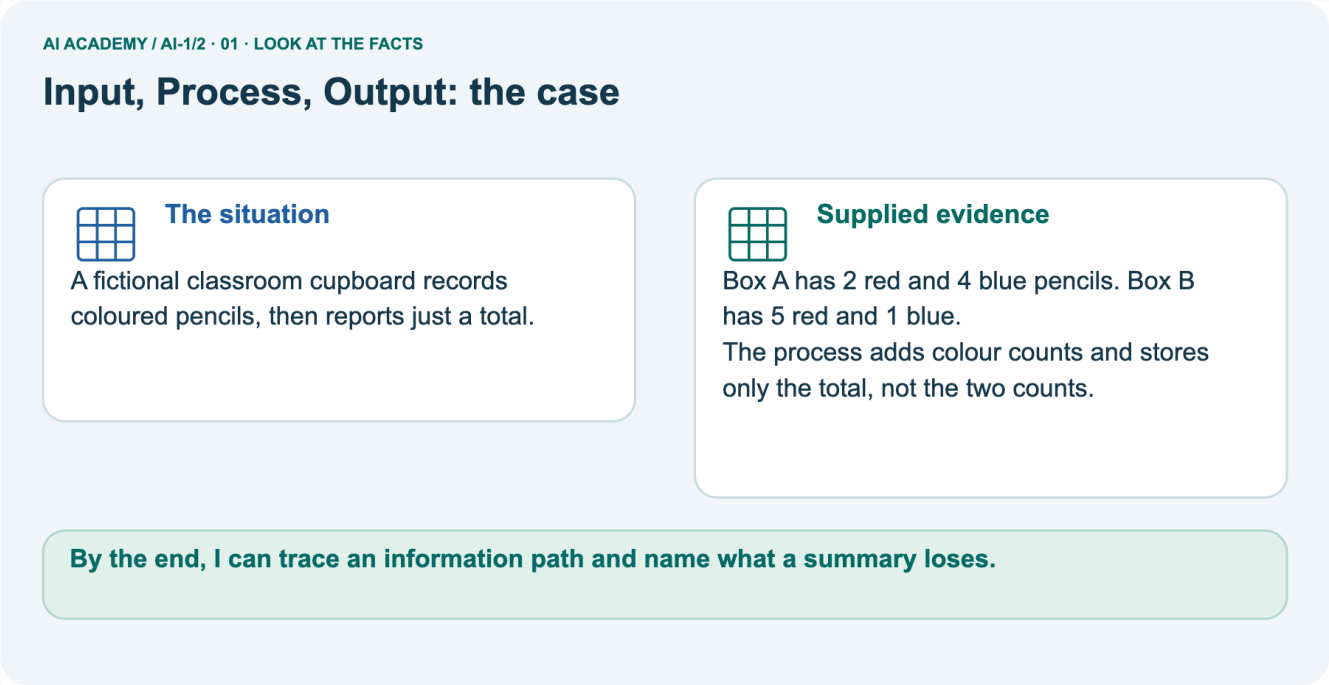
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Input, Process, Output picture from its labels, then record a relevant limit.

01. Input, Process, Output: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

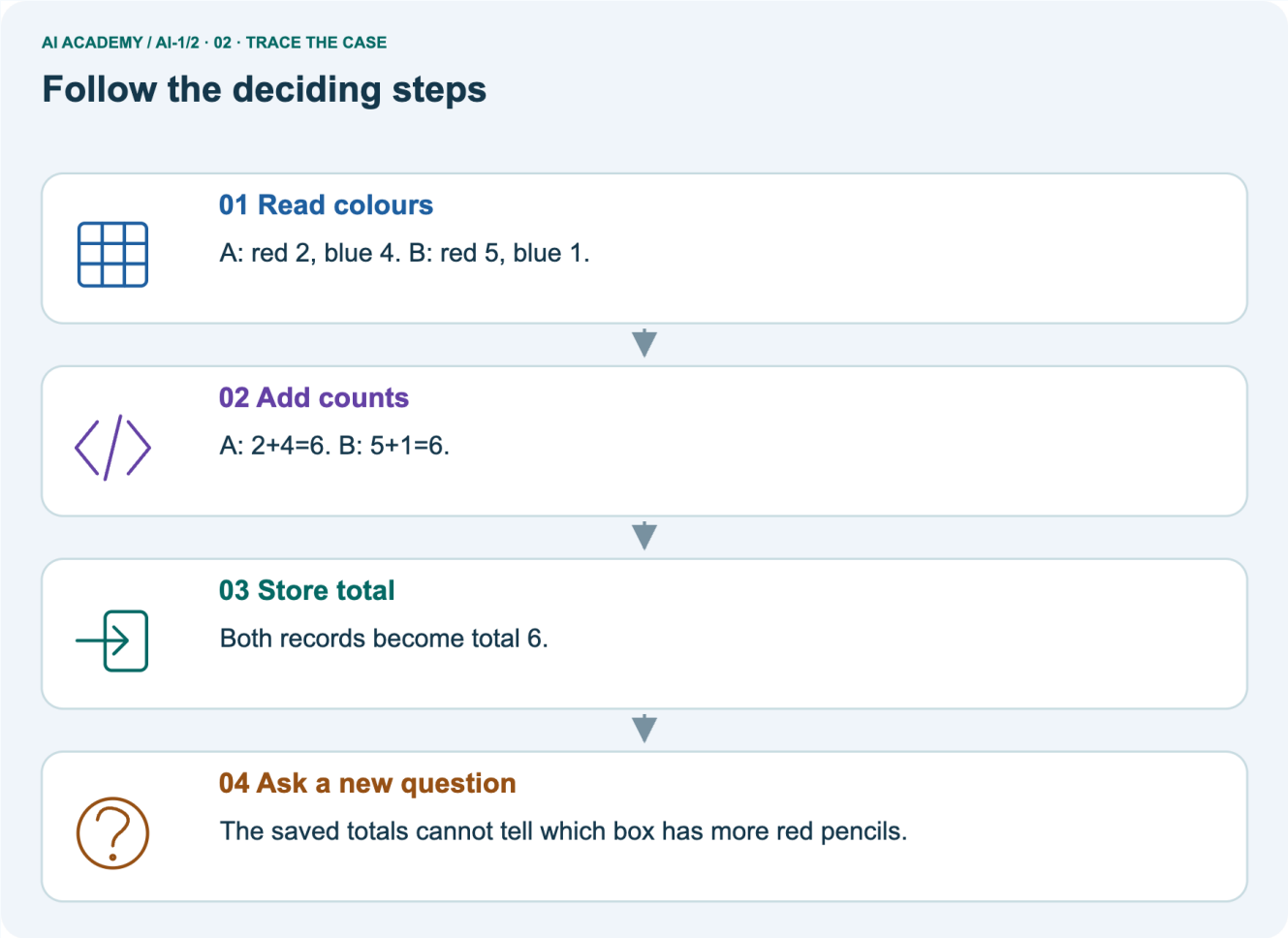
A fictional classroom cupboard records coloured pencils, then reports just a total. Supplied facts: Box A has 2 red and 4 blue pencils. Box B has 5 red and 1 blue.

The **process** adds colour counts and stores only the total, not the two counts.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



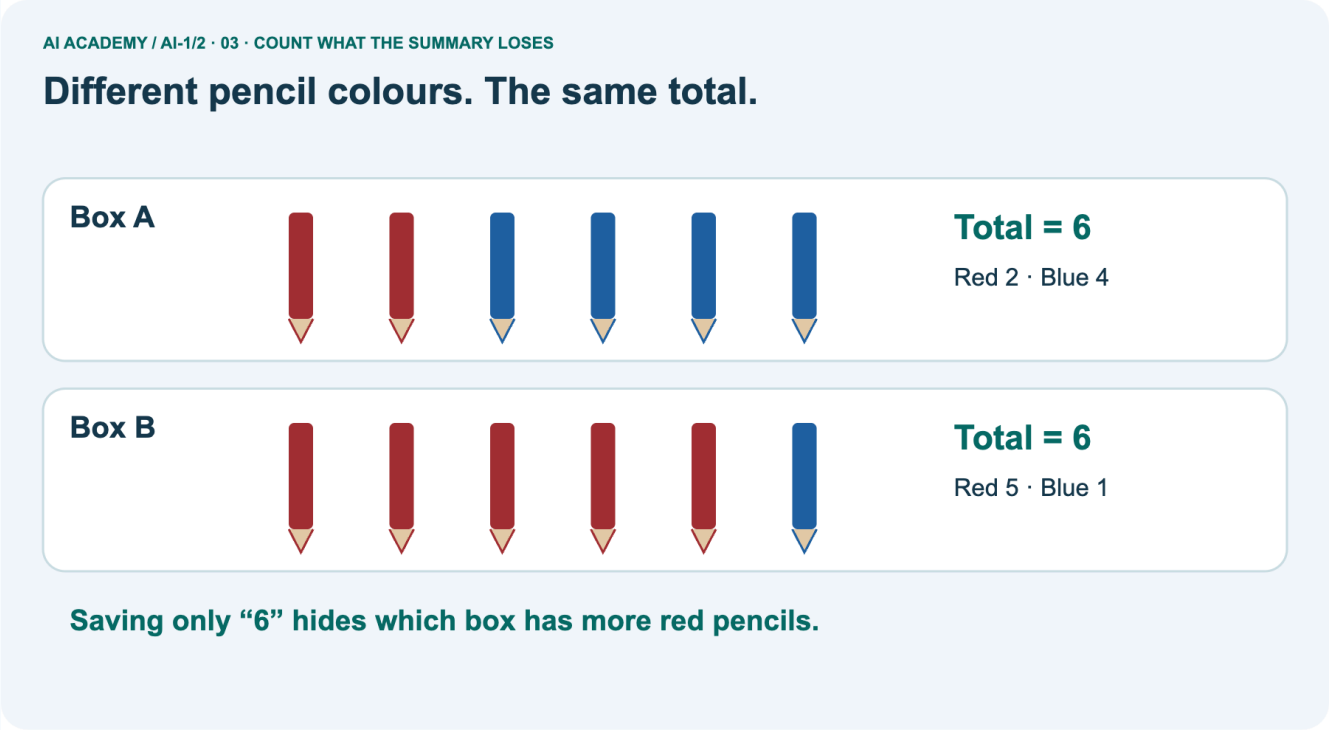
Follow the labelled picture with the matching explanation. Open the vector diagram

Read colours: A: red 2, blue 4. B: red 5, blue 1. → Add counts: A: 2+4=6. B: 5+1=6. → Store total: Both records become total 6. → Ask a new question: The saved totals cannot tell which box has more red pencils.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Different pencil colours. The same total.



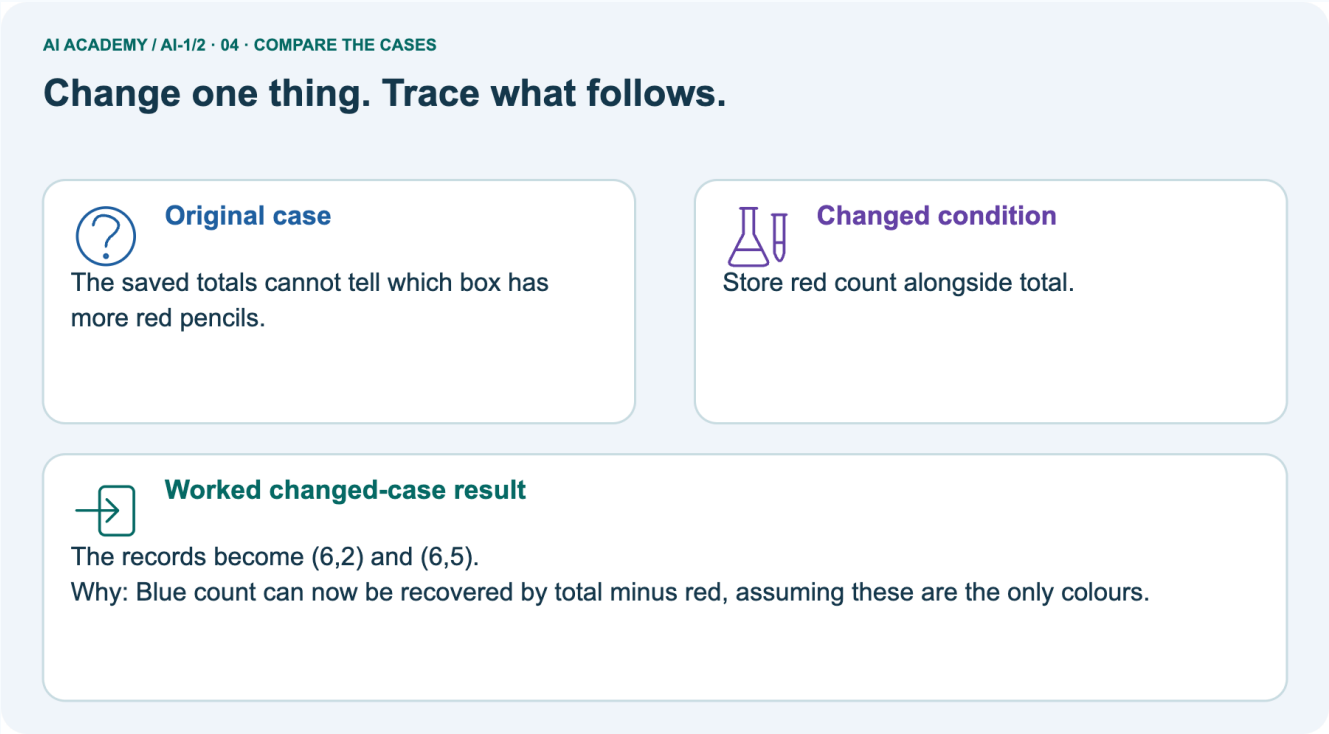
Follow the labelled picture with the matching explanation. Open the vector diagram

Box A: 2 red pencils and 4 blue, total 6. Box B: 5 red pencils and 1 blue, total 6. The equal totals do not preserve the colour counts.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The saved totals cannot tell which box has more red pencils. Changed condition: Store red count alongside total. Expected result: The records become (6,2) and (6,5).

Reason: Blue count can now be recovered by total minus red, assuming these are the only colours.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit

AI ACADEMY / AI-1/2 · 05 · EXPLAIN WITH CARE

Connect the explanation to its limit

What the case explains

Adding preserves the number of pencils but discards how that number was divided between colours. Both distinct inputs lead to the same stored output. Working backwards from 6 therefore has several possible answers; it is not an instruction to divide equally.

What it does not establish

The two-colour assumption matters. Extra colours would prevent inferring blue from total and red alone.

Follow the labelled picture with the matching explanation. Open the vector diagram

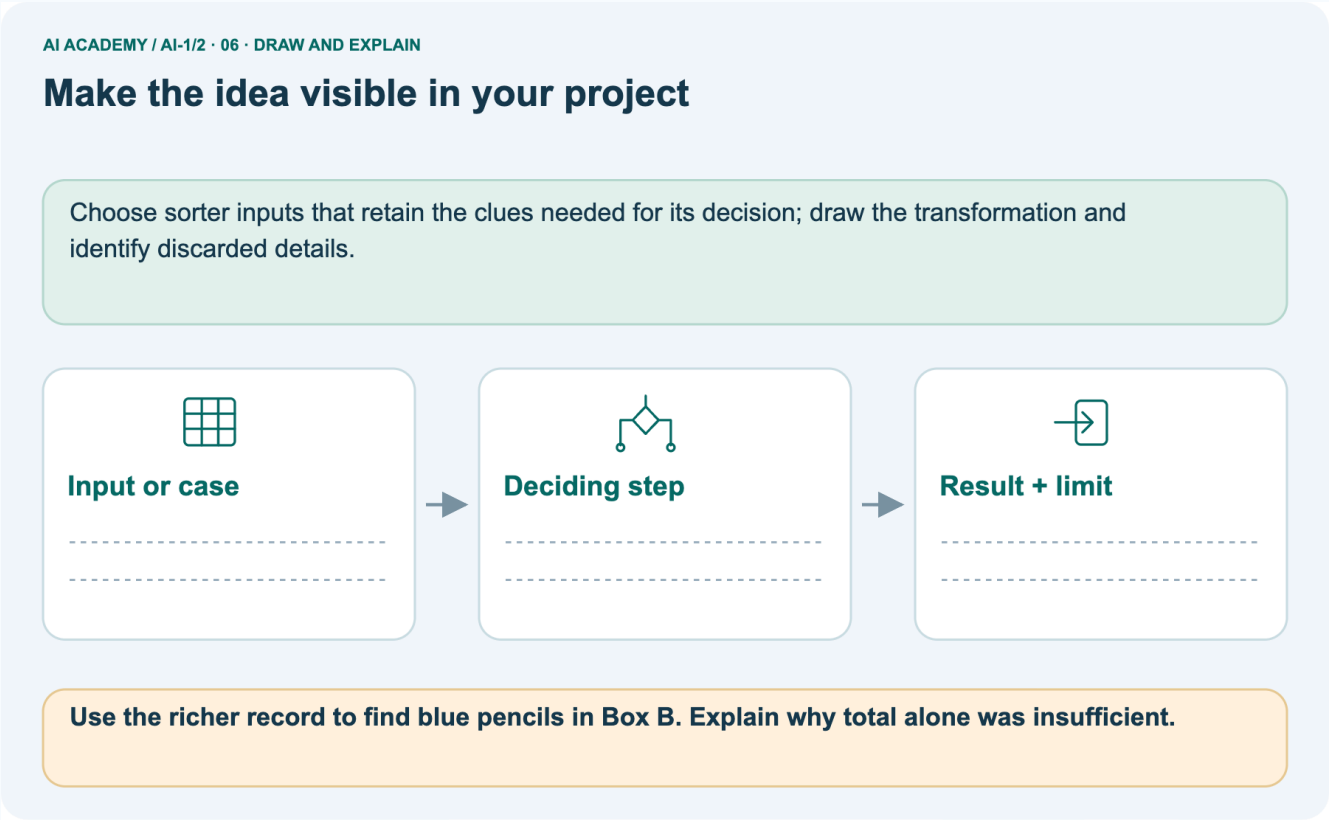
Adding preserves the number of pencils but discards how that number was divided between colours. Both distinct **input**s lead to the same stored **output**.

Working backwards from 6 therefore has several possible answers; it is not an instruction to divide equally. Limit: The two-colour assumption matters. Extra colours would prevent inferring blue from total and red alone.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Choose sorter **input**s that retain the clues needed for its decision; draw the transformation and identify discarded details. Draw your **input**, deciding step, result and limit.

Supported question: Use the richer record to find blue pencils in Box B. Explain why total alone was insufficient.

Draw your project **trace** and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.